SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Online K-Learning

Course: 61FIT3SE2 - Software Engineering 02 (Spring 2026)

Group: 11

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1.0. Introduction

1.1. Purpose

This SRS document has been prepared to establish a complete understanding of the K-Learn platform requirements. It serves to consolidate all functional and non-functional requirements gathered from stakeholders, including language learners, content administrators, and system managers. The document will guide the development team throughout the implementation phase and serve as a foundation for system testing and validation.

The primary objectives of this document are to:

- Present a comprehensive overview of the K-Learn platform, including its features, constraints, and capabilities
- Define the system's functional behavior and technical specifications
- Establish acceptance criteria for system validation
- Serve as a contractual agreement between stakeholders and the development team

This document is intended for developers, testers, project managers, system analysts, database administrators, and quality assurance personnel involved in the K-Learn project.

1.2. Scope of Project

K-Learn is a web-based Korean language learning platform designed to support students in achieving proficiency from beginner to advanced levels.

The system will provide interactive vocabulary lessons, comprehensive grammar instruction, quiz-based assessments, and personalized progress tracking.. The platform

will not include live tutoring sessions, mobile applications, or social networking features in its initial release.

The primary benefits of K-Learn include accessible self-paced learning, structured educational pathways, and data-driven progress monitoring. The platform aims to make Korean language education available to a global audience while providing educators with tools to manage content and track learner performance. K-Learn supports the growing demand for Korean language skills in business, education, and cultural exchange.

This specification aligns with the K-Learn Vision Document and the project's Business Requirements Documentation.

1.3. Glossary

Term	Definition
Hangul	The Korean alphabet writing system
TOPIK	Test of Proficiency in Korean - standardized exam with 6 levels
LMS	Learning Management System
SRS	Software Requirements Specification

1.4. References

The following documents are referenced in this SRS:

K-Learn Vision Document, Version 2.0, January 2026

K-Learn Project Plan, Version 1.5, January 2026

TOPIK Examination Standards, NIIED Korea, 2025

IEEE Standard for SRS (IEEE Std 830-1998)

1.5. Overview of Document

This document is divided into four main sections. Section 2 presents the overall description of K-Learn, including product perspective, general capabilities, user characteristics, operating environment, and design constraints. Section 3 details the specific functional and non-functional requirements, organized by feature sets and system attributes. Section 4 provides supporting information and appendices.
The SRS follows a hierarchical organization where high-level system descriptions in Section 2 are refined into detailed specifications in Section 3, ensuring traceability from business objectives to technical requirements.

2. GENERAL DESCRIPTION

2.1 Product Perspective

K-Learn is a web-based online Korean language learning platform designed to support self-paced learning through structured lessons and interactive exercises.
The system operates as a standalone application accessed via modern web browsers and does not require installation of additional client-side software beyond a standard browser.

K-Learn follows a client–server architecture, where users interact with the system through a web interface, while all business logic, data processing, and storage are handled on the server side. The platform integrates with external services such as speech-to-text APIs for speaking exercises and audio streaming services for listening exercises.

The product is part of a broader e-learning ecosystem and may later be extended to integrate with learning management systems (LMS) or mobile applications in future releases. However, such integrations are out of scope for the current version.

2.2 Product Functions

At a high level, K-Learn provides the following major functions:

- User authentication, including registration, login, and logout
- Lesson selection based on predefined learning paths
- Study of lesson theory content, including vocabulary and grammar topics
- Interactive practice through multiple exercise types:
    - Reading exercises
    - Listening exercises
    - Speaking exercises using speech recognition
    - Writing exercises with stroke-order guidance
- Lesson review to reinforce previously learned content

- Progress tracking and point accumulation
- Display of user rankings through a leaderboard
- Participation in online speaking rooms for real-time speaking practice

These functions are designed to support learners from beginner to advanced levels while maintaining consistency and reusability across lesson content.

2.3 User Characteristics

The primary users of K-Learn are language learners with varying levels of experience and technical proficiency. User characteristics include:

General Users (Learners)

- Basic computer and internet usage skills
- Familiarity with web-based applications
- No prior knowledge of the Korean language is required
- Motivation to learn Korean through self-paced study

Advanced Learners

- Higher expectations for content accuracy and progress tracking
- Will frequently use speaking and writing exercises

Administrators (Future Scope)

- Technical background sufficient to manage lesson content and user data
- Not included in the functional scope of the initial release

The system is designed to be intuitive and user-friendly so that users with minimal technical skills can effectively use all core features.

2.4 General Constraints

The development and operation of K-Learn are subject to the following constraints:

- The system must operate on standard web browsers (Chrome, Firefox, Edge, Safari)
- The initial release is limited to a web-based platform only; no mobile applications
- The system must comply with common web security standards, including secure authentication
- Speech recognition accuracy depends on the capabilities and limitations of third-party APIs
- Development must adhere to the course requirements of 61FIT3SE2 – Software Engineering 02

- Project timeline and features are constrained by academic semester deadlines and team resources

2.5 Assumptions and Dependencies

The following assumptions and dependencies are made during the development of K-Learn:

Assumptions

- Users have stable internet access
- Users possess devices with microphones for speaking exercises
- Users are willing to create an account to access full platform features

Dependencies

- Availability and reliability of third-party speech-to-text services
- Availability of audio streaming services for listening exercises
- Hosting infrastructure capable of supporting concurrent users
- Continued browser support for required web technologies (HTML5, JavaScript, WebRTC)
- Any changes to these assumptions or dependencies may impact system functionality and performance.

3.0. Requirements Specification

3.1 External Interface Requirements

The only link to an external system is the link to the Historical Society (HS) Database to verify the membership of a Reviewer. The Editor believes that a society member is much more likely to be an effective reviewer and has imposed a membership requirement for a Reviewer. The HS Database fields of interest to the Web Publishing Systems are member's name, membership (ID) number, and email address (an optional field for the HS Database).

The *Assign Reviewer* use case sends the Reviewer ID to the HS Database and a Boolean is returned denoting membership status. The *Update Reviewer* use case requests a list of member names, membership numbers and (optional) email addresses when adding a new Reviewer. It returns a Boolean for membership status when updating a Reviewer.

3.1.1 User Interfaces

The system shall provide a responsive and intuitive web-based interface (WBI) accessible via standard web browsers. The User Interface (UI) design must focus on readability and accessibility to support long study sessions.

- **General Layout:** The application shall utilize a consistent layout comprising a top navigation bar (Logo, Menu, User Profile/Points), a main content area, and a footer.
- **Responsive Design:** The UI shall adapt to various screen sizes (Desktop, Tablet) using a grid-based layout (e.g., Bootstrap or CSS Grid), ensuring functional parity across devices.
- **Key Screen Interfaces:**
    - **Dashboard:** Displays the learner's current level, recent course progress bars, and a summary of XP points/streaks.
    - **Lesson Player:** A specialized interface for taking lessons, divided into tabs or sequential steps: Theory, Reading, Listening (with audio player controls), Speaking (with recording button and visualizer), and Writing (with canvas or text input).

    - **Speaking Room:** A real-time communication interface featuring a list of active participants, microphone toggle controls, and status indicators (talking/muted).
- **Accessibility:** The UI shall support standard keyboard navigation and screen readers where possible, adhering to WCAG 2.1 Level A standards.

3.1.2 Hardware Interfaces

Since K-Learn is a web application, the hardware interfaces are primarily defined by the client-side devices used to access the system.

- **Client Devices:** The system shall support any personal computer (PC, Laptop) or tablet device capable of running a modern web browser.
- **Audio Input (Microphone):** The system requires a functional microphone (internal or external) on the client device to capture user voice data for **Speaking Exercises** (UC-08) and the **Online Speaking Room** (UC-12).
- **Audio Output (Speakers/Headphones):** The system requires audio output hardware to play pronunciation samples and listening comprehension tracks for **Listening Exercises** (UC-07).
- **Network Interface:** The client device must have an active network adapter (Wi-Fi or Ethernet) to connect to the K-Learn server.

3.1.3 Software Interfaces

The system interacts with various software components to deliver its functionality.

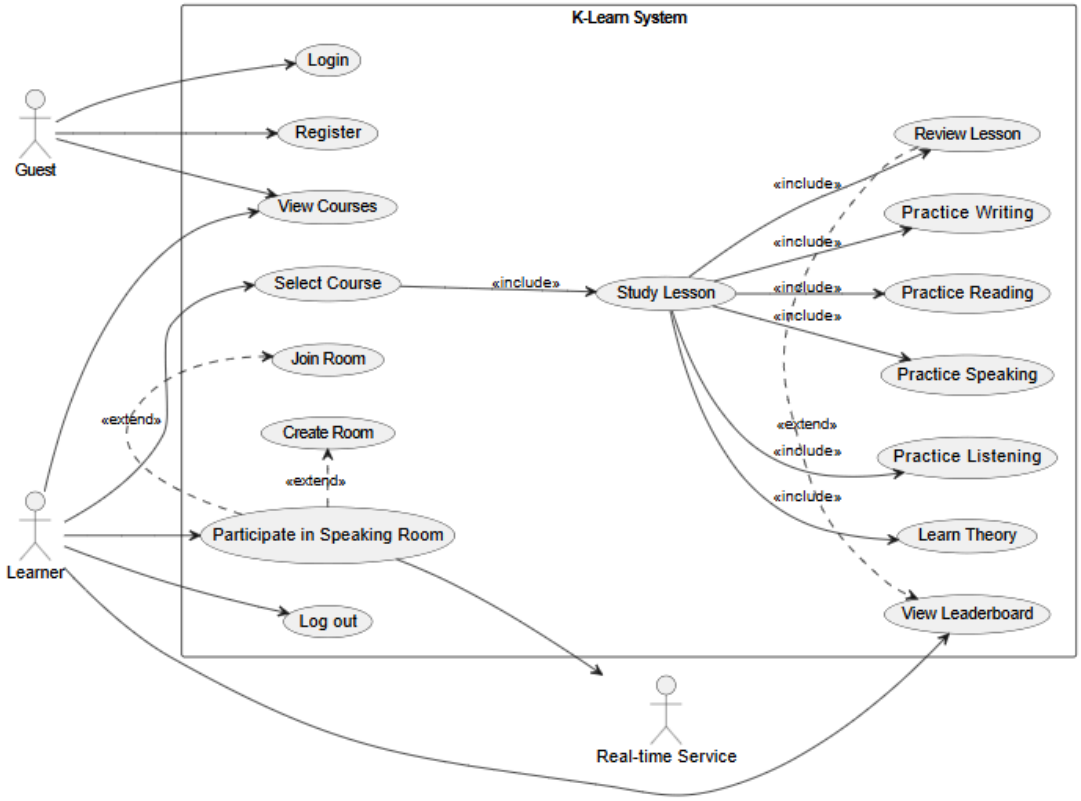
- **Web Browsers:** The system shall be compatible with the latest stable versions of major HTML5-compliant browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.
- **Database Management System (DBMS):** The application server shall interface with a relational database (e.g., MySQL or PostgreSQL) to store user profiles, lesson content, and progress logs.
- **Speech-to-Text (STT) Engine:** The system shall interface with an external Speech Recognition API (e.g., Web Speech API or a third-party cloud service like Google Cloud Speech-to-Text) to convert user voice input into text for accuracy evaluation.
- **Operating Systems:** The client-side application shall be OS-agnostic, functioning on Windows, macOS, Linux, Android, and iOS via the browser.

3.1.4 Communications Interfaces

The system shall use standard network protocols to ensure secure and efficient data transmission.

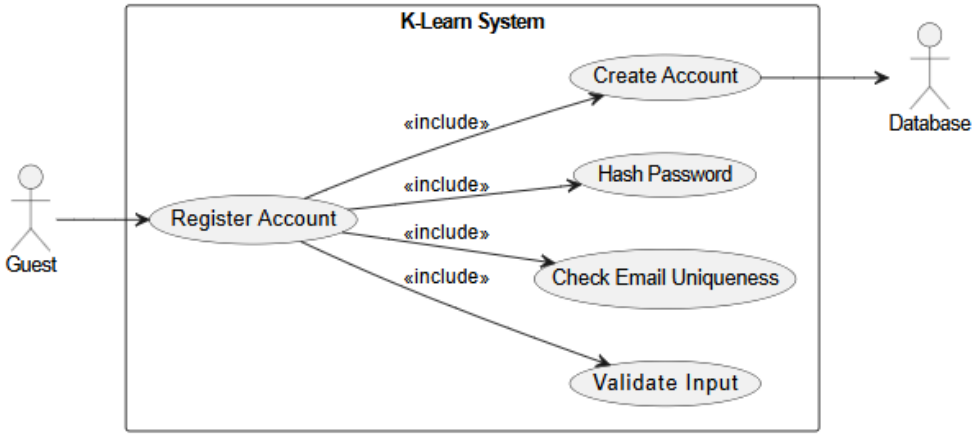
- **Communication Protocols:**
    - **HTTPS (Hypertext Transfer Protocol Secure):** All client-server communication shall be encrypted using TLS/SSL (HTTPS) to protect user credentials and data.
    - **WSS (WebSockets Secure) / WebRTC:** The system shall use secure WebSockets or WebRTC protocols to facilitate low-latency, real-time audio transmission for the **Online Speaking Room** (UC-12).
- **Data Exchange Format:**
    - **JSON (JavaScript Object Notation):** The system shall use JSON for structured data exchange between the client (Frontend) and the server (Backend API), including lesson content delivery and quiz result submission.
- **API Standards:** The backend shall expose a RESTful API to handle requests from the web client (e.g., `POST /api/login`, `GET /api/lessons/{id}`).

3.2 Functional Requirements



3.2.1 Guest

3.2.1.1 Register

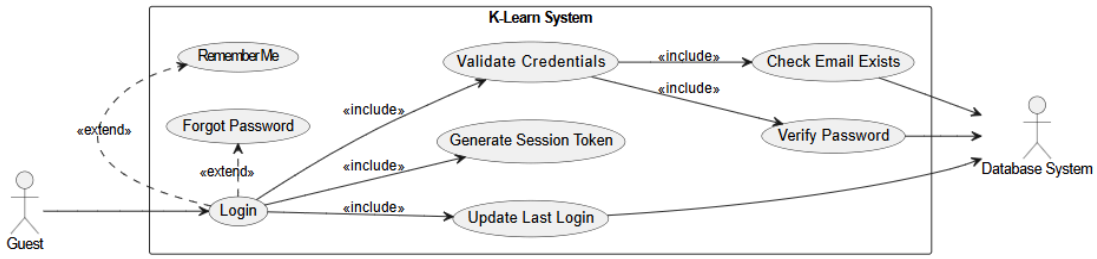


Id& Name	UC-02 Register

Description	This use case allows a guest to create a new user account in the K-Learn system. The process includes collecting user information, validating input data, creating an account in the database, and initializing a learning profile so the user can begin lessons and accumulate points.
Trigger	The guest clicks the "Sign Up" button on the Login screen.
Primary Actor	Guest
Secondary Actor	Database Management System
Basic Path	1. The system displays the **Registration Form** requesting: Full Name, Email Address, Password, and Confirm Password. 2. The user enters the required information and clicks "Create Account". 3. The system validates the **Email** format (using standard Regex). 4. The system checks the database to ensure the Email is unique. 5. The system validates **Password** strength (min 8 chars, mixed case, numbers). 6. The system checks if **Password** and **Confirm Password** match. 7. The system hashes the password before storage (e.g., using Bcrypt). 8. The system creates a new record in the Users table. 9. The system automatically initializes a Learner_Progress record linked to the new account, setting defaults: Current_Level = 1, Total_Points = 0. 10. The system displays a "Registration Successful" message and redirects the user to the Login screen.
Alternative Paths	**A1: Auto-Login after Registration** 1. In step 10, instead of redirecting to Login, the system automatically creates a session. 2. The system redirects the user directly to the **Dashboard** or **Select Lesson** page. **A2: Cancel Registration** 1. At any step before step 8, the user clicks "Cancel". 2. The system discards input data and returns to the previous page.
Pre-condition	The user is not logged into the system. The system and database are operational.
Post-condition	A new user account is created in the database. A Learning Profile (Level 1, 0 Points) is ready. The user can log in using the registered credentials.
Exception Paths	**E1: Duplicate Email**

	1. In step 4, the system detects the Email already exists. 2. The system displays an error: "Email already in use. Please login." **E2: Weak Password** 1. In step 5, the password does not meet security policies. 2. The system displays requirements: "Password must be 8+ chars with letters and numbers." **E3: Database Error** 1. In step 8 or 9, the system fails to write data. 2. The system displays a generic error: "System error. Please try again later."
Priority	Critical

3.2.1.2 Login

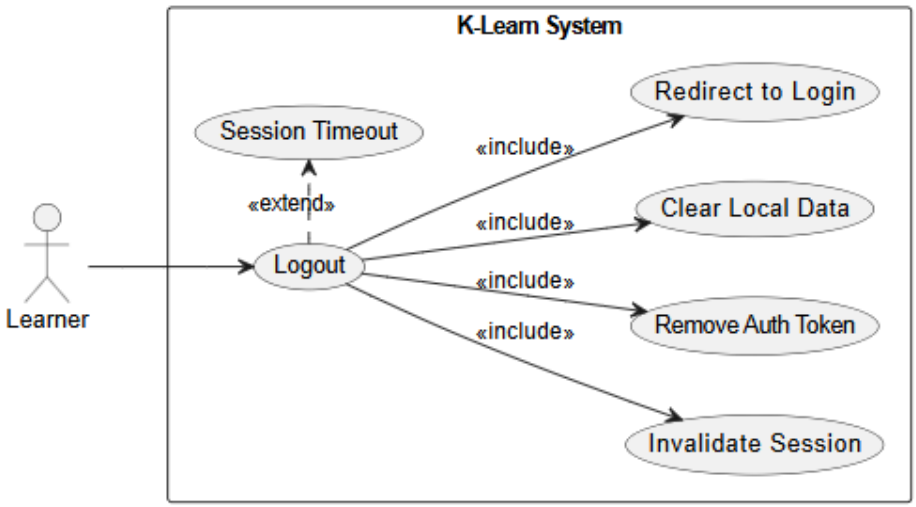


Id& Name	**UC-01 Login**
Description	This use case describes how a registered user authenticates to access protected learning resources. Upon success, the system grants access to lessons, exercises, and the leaderboard, while establishing a session to track progress.
Trigger	The user opens the K-Learn website (if not logged in) or clicks "Login".
Primary Actor	Guest
Secondary Actor	Identity Provider / Database System
Basic Path	1. The system displays the **Login Interface** with Email and Password fields. 2. The user enters their registered Email and Password. 3. The user clicks "Login". 4. The system sends an authentication request via HTTPS. 5. The system verifies the existence of the Email in the database. 6. The system hashes the input password and compares it with the stored hash. 7. If they match, the system generates a **Session Token** (e.g., JWT) containing user identity (user_id, role).

	8. The system updates the last_login_at timestamp in the user profile. 9. The system redirects the user to the **Dashboard**.
Alternative Paths	**A1: "Remember Me"** 1. In step 2, the user checks the "Remember Me" box. 2. In step 7, the system sets a longer expiration time for the Token (e.g., 30 days). **A2: Forgot Password** 1. The user clicks "Forgot Password?". 2. System redirects to the Password Recovery flow.
Pre-condition	The user already has a registered account. The system and authentication services are available.
Post-condition	The user is authenticated and can access internal functions. Auth Token is stored securely (Cookie/LocalStorage). UI updates to show User Name and Avatar.
Exception Paths	**E1: Invalid Credentials** 1. Email not found or Password incorrect. 2. System returns error code AUTH_INVALID. 3. System displays: "Invalid email or password." **E2: Account Locked** 1. User enters wrong password 5 times. 2. System locks account for 15 minutes.
Priority	**Critical**

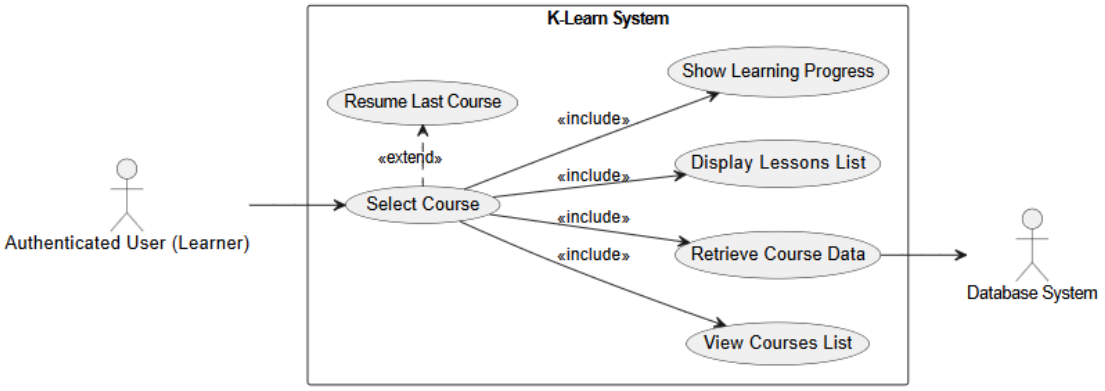
3.2.2 Learner

3.2.2.1 Logout



Id& Name	UC-03 – Log out
Description	This use case allows a logged-in user to securely terminate their session. The system invalidates authentication tokens and clears temporary session data on the browser to prevent unauthorized access on the same device.
Trigger	User clicks "Logout" from the profile menu or navigation bar.
Primary Actor	User
Secondary Actor	None
Basic Path	1. The user clicks **"Logout"**. 2. The system sends a logout request to the server. 3. The server invalidates the current session (or adds the token to a blacklist). 4. The client removes the authentication token (e.g., cookie or local storage). 5. The system clears any local session-related data. 6. The system redirects the user to the Home page 7. The system displays a confirmation message: *"You have been logged out."*
Alternative Paths	**A1: Session Timeout** 1. The user is inactive for a specific period (e.g., 30 mins). 2. Token expires. 3. Upon next action, the system detects invalid token. 4. The system performs Logout (Steps 4-6 of Basic Path). 5. System redirects to Login with message: "Session expired."
Preconditions	The user is currently authenticated (logged in). A valid session or token exists.
Postcondition	- The session is terminated. - Authentication tokens are invalidated. - Protected resources are no longer accessible. - User-specific data is cleared from the client side.
Exception Paths	**E1: Network Error During Logout** 1. The logout request fails due to network issues. 2. The system forces client-side token removal to protect the user. 3. The system redirects the user to the Login screen as if logout was successful.
Priority	High

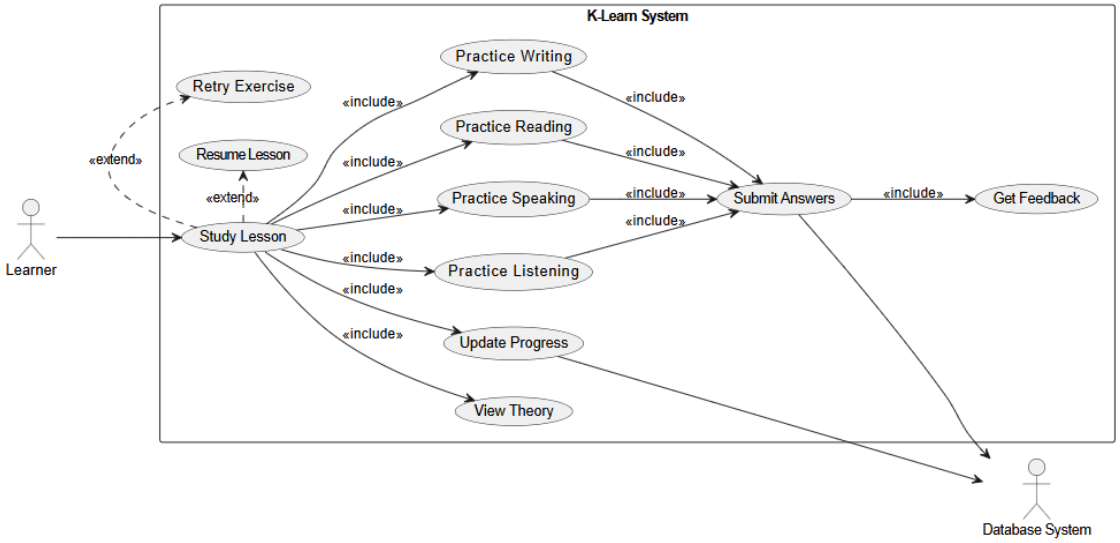
3.2.2.2 Select Course



Id& Name	UC-04 – Select Course
Description	This use case allows an authenticated user to browse and select a course (e.g., Beginner 1, Beginner 2) in order to access its associated lessons.
Trigger	The user navigates to the Courses page or clicks on a course selection option.
Primary Actor	Learner
Secondary Actor	Database System
Basic Path	1. The system displays a list of available courses (e.g., Beginner 1, Beginner 2). 2. The user browses the list of courses. 3. The user selects a course. 4. The system retrieves course details and associated lessons from the database. 5. The system displays the list of lessons within the selected course in sequential order. 6. The system highlights the user's current progress (if any).
Alternative Paths	**A1: Resume Last Course** 1. The system detects the user has previous progress. 2. The system suggests resuming the last accessed course. 3. The user chooses to continue or select a different course.
Preconditions	- The user is logged into the system. - Courses are available in the system database.
Postcondition	- The selected course is loaded. - The user can view and access lessons within the course. - The system is ready for the next action (e.g., Study Lesson).
Exception Paths	**E1: No Courses Available** 1. The system finds no available courses. 2. The system displays a message: *"No courses available at the moment."* **E2: Data Retrieval Error** 1. The system fails to load course data from the database.

Id& Name	UC-04 – Select Course
	2. The system displays: *"Unable to load courses. Please try again later."*
Priority	High

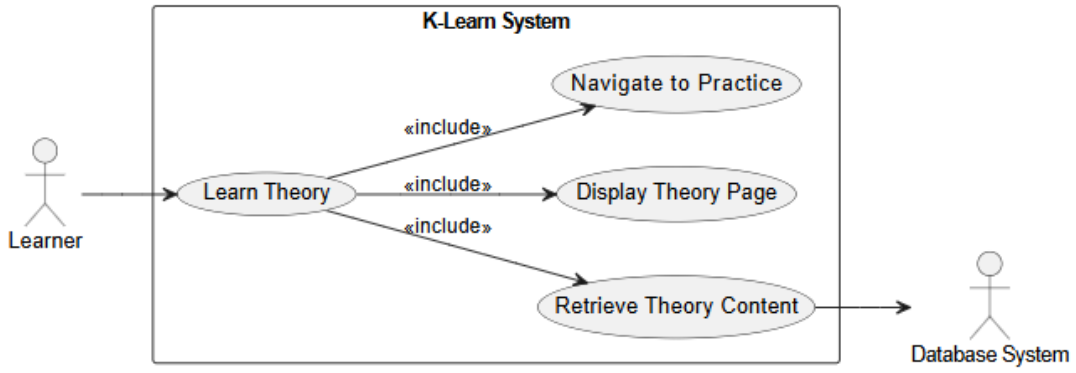
3.2.2.3 Select Lesson



ID & Name	UC-05 – Study Lesson
Description	This use case allows an authenticated user to study a selected lesson, including theory content and practice activities such as listening, speaking, reading, and writing exercises.
Trigger	The user selects a lesson from a course.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system. - A course has been selected.- The lesson exists in the system.

Basic Flow	1. The system displays the lesson content. 2. The user views the theory section of the lesson. 3. The user selects a practice section (Listening, Speaking, Reading, or Writing). 4. The system loads the selected practice content. 5. The user completes exercises in the selected section. 6. The system evaluates the user's answers (if applicable). 7. The system provides feedback and/or correct answers. 8. The system updates the user's learning progress. 9. The user may continue with another section or move to the next lesson.
Alternative Flows	**A1: Skip Section** 1. The user skips a section (e.g., does not complete Listening). 2. The system allows the user to continue with another section or next lesson. **A2: Resume Lesson** 1. The user previously started the lesson. 2. The system restores the last completed section and progress. **A3: Retry Exercise** 1. After receiving feedback, the user chooses to retry the exercise. 2. The system reloads the exercise.
Postconditions	- The user's progress is updated. - Completed sections are recorded. - The system is ready for the next learning action (next section or next lesson).
Exception Flows	**E1: Failed to Load Lesson Content** 1. The system cannot retrieve lesson data. 2. The system displays: *"Unable to load lesson. Please try again later."* **E2: Submission Error** 1. The system fails to submit user answers. 2. The system prompts the user to retry.
Priority	Critical

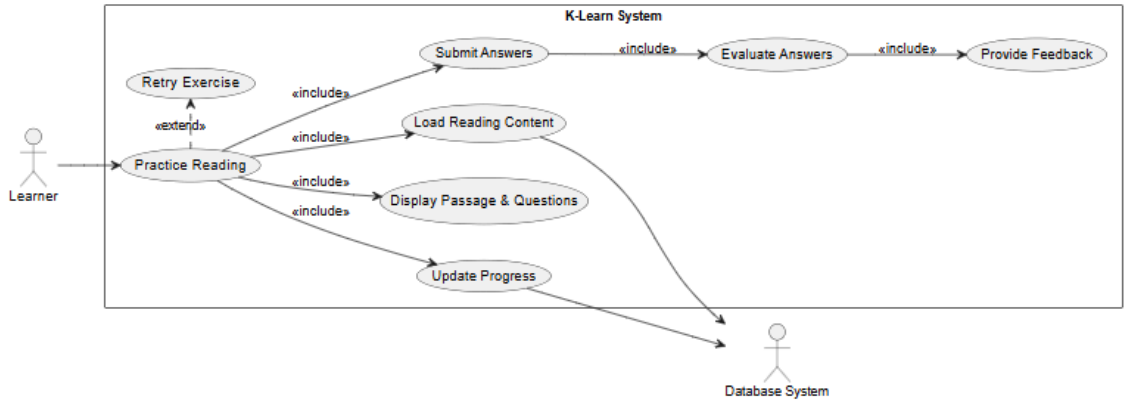
3.2.2.4 Learn Lesson Theory



ID & Name	UC-06 – Learn Theory
Description	This use case allows an authenticated user to view the theory content of a selected lesson. The content is presented on a single page, covering all theoretical knowledge required before starting practice activities.
Trigger	The user selects the “Theory” section within a lesson.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system.- A lesson has been selected.- The lesson contains theory content.
Basic Flow	1. The system retrieves the theory content of the selected lesson from the database. 2. The system displays the full theory content on a single page. 3. The user reads and reviews the theory. 4. The system provides a **“Start Practice”** button. 5. The user clicks **“Start Practice”**. 6. The system navigates the user to the practice section of the lesson.
Alternative Flows	**A1: Skip Theory** 1. The user does not read the content. 2. The user directly clicks **“Start Practice”**. **A2: Revisit Theory**

	1. The user has previously viewed the theory. 2. The system reloads the same theory page.
Postconditions	- The theory content has been displayed to the user. - The user proceeds to the practice section. - The system may record that the theory has been viewed.
Exception Flows	**E1: Failed to Load Theory Content** 1. The system cannot retrieve theory data. 2. The system displays: *"Unable to load content. Please try again later."*
Priority	High

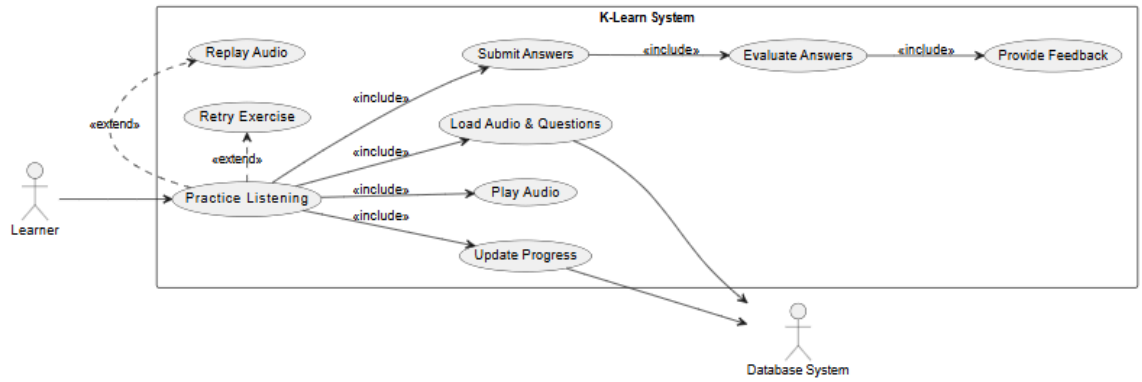
3.2.2.5 Practice Reading



ID & Name	UC-07 – Practice Reading
Description	This use case allows an authenticated user to practice reading skills within a lesson. The user completes reading exercises such as multiple-choice and fill-in-the-blank, and receives feedback on their performance.
Trigger	The user selects the **Reading** section within a lesson.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system.- A lesson has been selected.- The lesson contains reading exercises.

Basic Flow	1. The system retrieves reading exercises from the database. 2. The system displays reading content (passage and questions). 3. The user reads the content and answers the questions. 4. The user submits their answers. 5. The system evaluates the answers. 6. The system displays feedback and correct answers. 7. The system updates the user's progress and score.
Alternative Flows	**A1: Retry Exercise** 1. After receiving feedback, the user chooses to retry. 2. The system reloads the exercise. **A2: Skip Exercise** 1. The user skips the reading section. 2. The system allows moving to another section.
Postconditions	- The user's answers are recorded. - Feedback is provided. - Progress is updated.
Exception Flows	**E1: Failed to Load Exercises** 1. The system cannot retrieve data. 2. The system displays: *"Unable to load exercises. Please try again later."* **E2: Submission Error** 1. The system fails to submit answers. 2. The system prompts the user to retry.
Priority	High

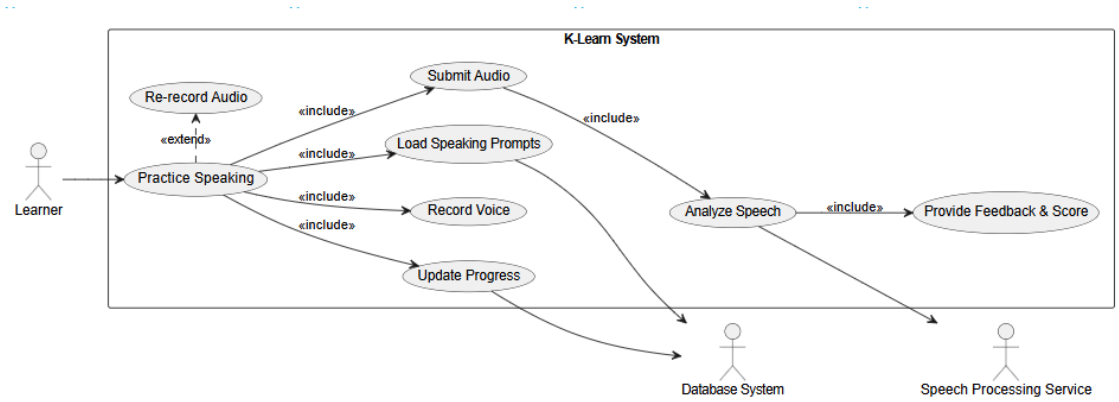
3.2.2.6 Practice Listening



ID & Name	UC-08 – Practice Listening
Description	This use case allows an authenticated user to practice listening skills by interacting with audio-based exercises. The user listens to audio content, answers related questions, and receives feedback on their performance.
Trigger	The user selects the **Listening** section within a lesson.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system. - A lesson has been selected. - The lesson contains listening exercises with audio content.
Basic Flow	1. The system retrieves listening exercises and audio files from the database. 2. The system displays the listening interface with audio controls (play, pause, replay). 3. The user plays the audio. 4. The user listens and answers the questions. 5. The user submits their answers. 6. The system evaluates the answers. 7. The system displays feedback and correct answers. 8. The system updates the user's progress and score.
Alternative Flows	**A1: Replay Audio** 1. The user clicks replay. 2. The system plays the audio again. **A2: Retry Exercise** 1. After feedback, the user chooses to retry. 2. The system reloads the exercise. **A3: Skip Exercise** 1. The user skips the listening section. 2. The system allows moving to another section.
Postconditions	- The user's answers are recorded. - Feedback is provided. - Progress is updated.

Exception Flows	**E1: Audio Load Failure** 1. The system fails to load audio. 2. The system displays: *"Unable to load audio. Please try again later."* **E2: Submission Error** 1. The system fails to submit answers. 2. The system prompts the user to retry.
Priority	High

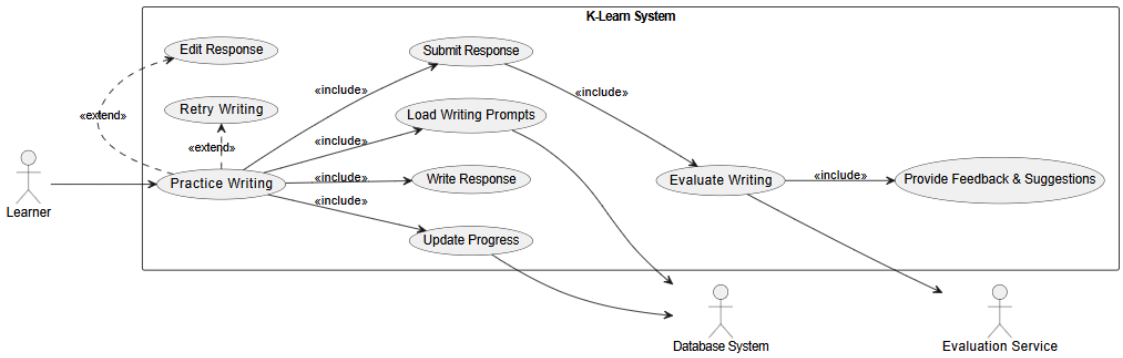
3.2.2.7 Practice Speaking



ID & Name	UC-09 – Practice Speaking
Description	This use case allows an authenticated user to practice speaking skills by recording their voice in response to prompts. The system analyzes pronunciation and provides feedback and scoring.
Trigger	The user selects the **Speaking** section within a lesson.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Speech Processing Service / Database System
Preconditions	- The user is logged into the system. - A lesson has been selected. - The lesson contains speaking exercises. - The user grants microphone access.

Basic Flow	1. The system retrieves speaking prompts from the database. 2. The system displays the speaking interface with recording controls. 3. The user clicks **"Start Recording"**. 4. The system captures the user's voice input. 5. The user clicks **"Stop Recording"**. 6. The system sends the audio to the speech processing service. 7. The system analyzes pronunciation and fluency. 8. The system generates a score and feedback. 9. The system displays feedback and scores to the user. 10. The system updates the user's progress.
Alternative Flows	**A1: Re-record Audio** 1. The user is not satisfied with the recording. 2. The user records again. **A2: Skip Exercise** 1. The user skips the speaking section. 2. The system allows moving to another section.
Postconditions	- The user's speaking attempt is recorded. - Feedback and score are provided. - Progress is updated.
Exception Flows	**E1: Microphone Access Denied** 1. The user denies microphone permission. 2. The system displays: *"Microphone access is required for speaking practice."* **E2: Audio Processing Error** 1. The system fails to process the audio. 2. The system prompts the user to retry.
Priority	High

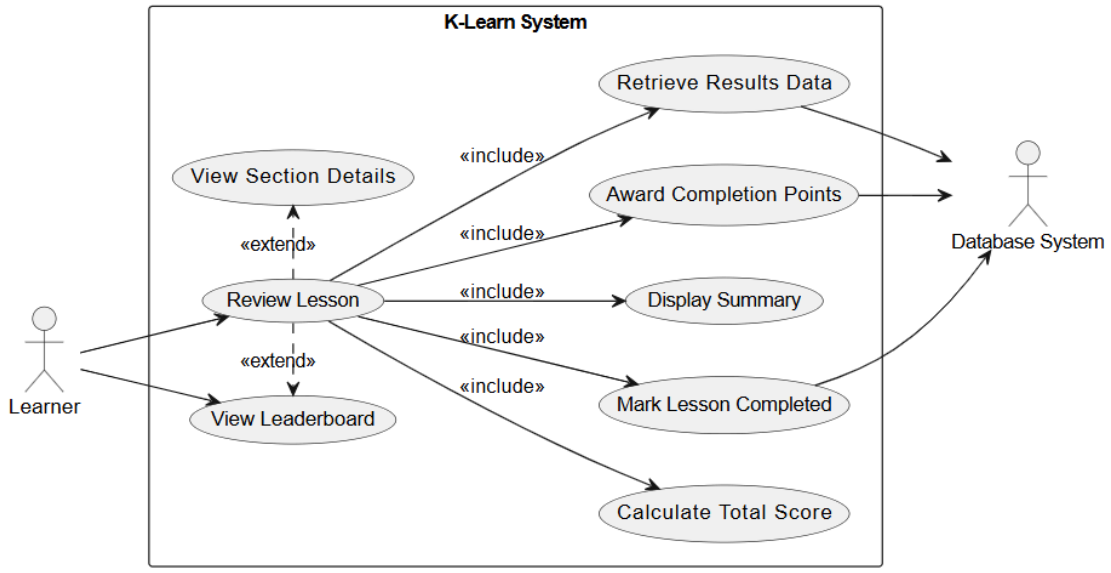
3.2.2.8 Practice Writing



ID & Name	UC-10 – Practice Writing
Description	This use case allows an authenticated user to practice writing skills by composing text responses based on prompts. The system evaluates the submission and provides feedback and scoring.
Trigger	The user selects the **Writing** section within a lesson.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Evaluation Service / Database System
Preconditions	- The user is logged into the system.- A lesson has been selected.- The lesson contains writing exercises.
Basic Flow	1. The system retrieves writing prompts from the database. 2. The system displays the writing interface with input fields. 3. The user reads the prompt and enters their response. 4. The user submits their answer. 5. The system sends the response to the evaluation service. 6. The system analyzes grammar, vocabulary, and structure. 7. The system generates a score and feedback. 8. The system displays feedback and suggestions for improvement. 9. The system updates the user's progress.
Alternative Flows	**A1: Edit Before Submission** 1. The user modifies their answer before submitting. **A2: Retry Writing** 1. After feedback, the user chooses to rewrite. 2. The system allows resubmission.

	A3: Skip Exercise 1. The user skips the writing section. 2. The system allows moving to another section.
Postconditions	- The user's writing submission is stored. - Feedback and score are provided. - Progress is updated.
Exception Flows	**E1: Submission Error** 1. The system fails to submit the response. 2. The system prompts the user to retry. **E2: Evaluation Service Error** 1. The system fails to evaluate the response. 2. The system displays: *"Unable to evaluate your answer. Please try again later."*
Priority	High

3.2.2.9 Review Lesson

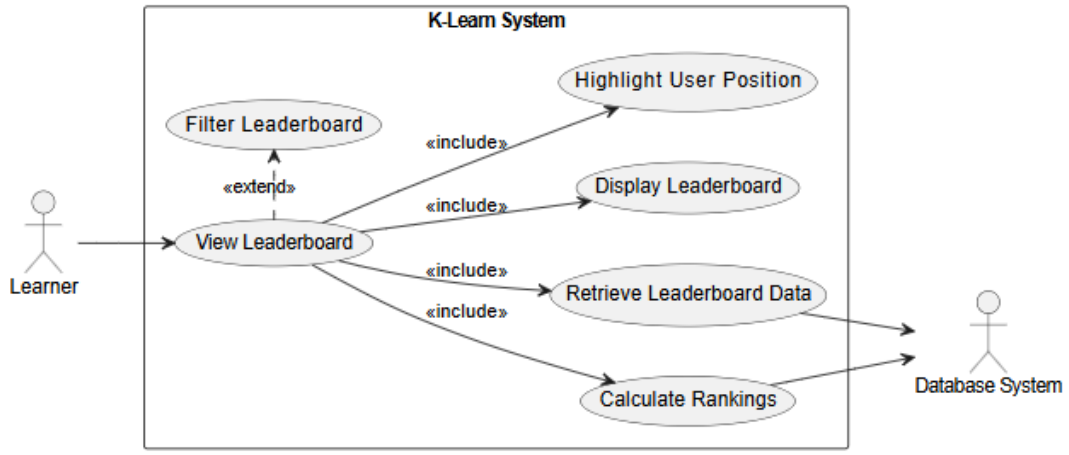


ID & Name	UC-11 – Review Lesson
Description	This use case allows an authenticated user to review their overall performance after completing a lesson. The system presents a

	summary including total score, section-wise results, and completion status.
Trigger	The user completes all required sections of a lesson or selects the **"Review"** option.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system. - The lesson has been completed (all required sections finished).
Basic Flow	1. The system verifies that all required lesson sections are completed. 2. The system calculates the overall lesson score. 3. The system retrieves detailed results from the database. 4. The system displays a summary page including: ● Total score ● Scores per section (Listening, Speaking, Reading, Writing) ● Completion status 5. The user reviews their performance. 6. The system awards completion points. 7. The system marks the lesson as completed.
Alternative Flows	**A1: View Section Details** 1. The user selects a specific section (e.g., Reading). 2. The system displays detailed results for that section. **A2: View Leaderboard** 1. The user selects **"View Leaderboard"**. 2. The system redirects to the leaderboard page.
Postconditions	- The lesson is marked as completed. - Scores and completion points are stored. - User progress is updated.
Exception Flows	**E1: Lesson Not Completed** 1. The system detects incomplete sections. 2. The system displays: *"Complete all sections to view the review."* **E2: Data Retrieval Error** 1. The system fails to load results. 2. The system displays: *"Unable to load review data. Please try again later."*

Priority	High

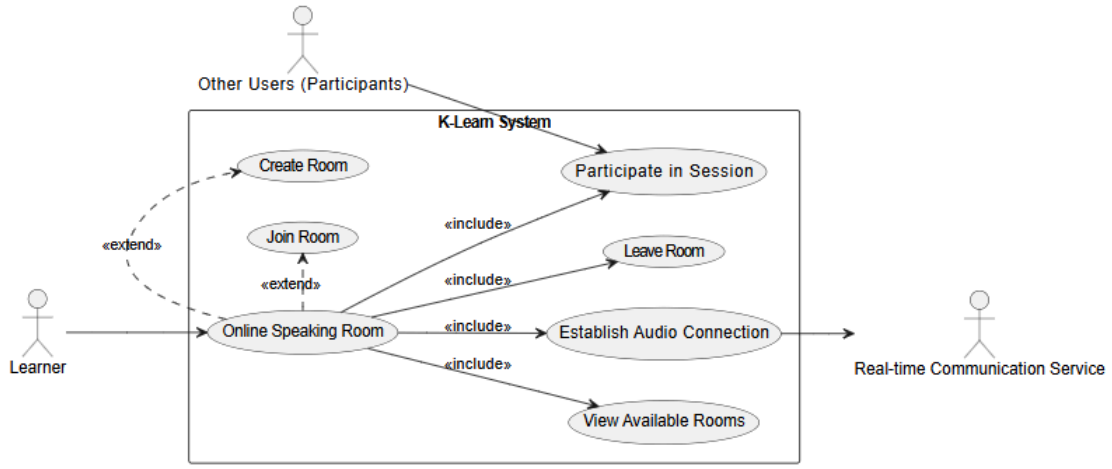
3.2.2.10 View Leaderboard



ID & Name	UC-12 – View Leaderboard
Description	This use case allows an authenticated user to view the leaderboard and compare their performance with other learners based on scores, rankings, or points.
Trigger	The user selects the **"Leaderboard"** option (from navigation or after reviewing a lesson).
Primary Actor	Authenticated User (Learner)
Secondary Actor	Database System
Preconditions	- The user is logged into the system. - Leaderboard data is available.
Basic Flow	1. The system retrieves leaderboard data from the database. 2. The system calculates rankings based on scores or points. 3. The system displays the leaderboard including: ● User rankings ● User names / avatars ● Scores or points 4. The system highlights the current user's position. 5. The user views and compares their ranking with others.

Alternative Flows	**A1: Filter Leaderboard** 1. The user selects a filter (e.g., by course, level, or time period). 2. The system updates the leaderboard accordingly. **A2: View from Review Lesson** 1. The user selects **"View Leaderboard"** from the Review page. 2. The system redirects to the leaderboard view.
Postconditions	- The leaderboard is displayed. - The user understands their ranking and relative performance.
Exception Flows	**E1: No Data Available** 1. The system finds no leaderboard data. 2. The system displays: *"No leaderboard data available."* **E2: Data Retrieval Error** 1. The system fails to load leaderboard data. 2. The system displays: *"Unable to load leaderboard. Please try again later."*
Priority	Medium

3.2.2.11 Online Speaking Room



Field	Content
ID & Name	UC-13 – Participate in Speaking Room

Description	This use case allows an authenticated user to create or join a real-time online speaking room to practice Korean speaking skills with other users.
Trigger	The user selects the **"Online Speaking Room"** feature.
Primary Actor	Authenticated User (Learner)
Secondary Actor	Other Users (Participants), Real-time Communication Service
Preconditions	- The user is logged into the system. - Microphone permission is granted. - Network connection is available.
Basic Flow	1. The user selects the Online Speaking Room feature. 2. The system displays a list of available speaking rooms. 3. The user chooses to: ● Create a new room, or ● Join an existing room. 4. The system establishes a real-time audio connection. 5. The user participates in the speaking session with other users. 6. The user leaves the room. 7. The system terminates the connection and returns the user to the main interface.
Alternative Flows	**A1: Create Room** 1. The user selects **"Create Room"**. 2. The user enters a room name. 3. The system creates the room and connects the user. **A2: Join Room** 1. The user selects an existing room. 2. The system connects the user to that room. 3. The system displays current participants. **A3: Leave Early** 1. The user leaves the room before the session ends. 2. The system disconnects the user.
Postconditions	- The user has joined and/or participated in a speaking session. - The user has exited the room successfully. - Lesson progress data remains unchanged.

Exception Flows	**E1: Microphone Permission Denied** 1. The user denies microphone access. 2. The system prompts the user to enable microphone permission. **E2: Connection Failure** 1. The system fails to establish a real-time connection. 2. The system displays an error and allows retry. **E3: Room Capacity Reached** 1. The selected room is full. 2. The system notifies the user and suggests another room.
Priority	Medium

3.3 Performance Requirements

The system shall satisfy the following performance targets (baseline):

REQ#	Priority	Description	Rationale	Use Cases
NFR-01	Critical	The system shall respond to at least 95% of user requests within 1 second under normal operating conditions.	To ensure a smooth and responsive learning experience for users.	UC-04 Select Course, UC-05 Study Lesson, UC-07 Practice Reading
NFR-02	High	The system shall provide a user-friendly interface and support localization for multiple languages.	To improve usability and accessibility for diverse learners.	All user-facing use cases
NFR-03	Medium	The system shall maintain a minimum uptime of 99.9%, excluding scheduled maintenance periods.	To ensure continuous availability of the learning platform.	All use cases

3.6 Other Requirements

3.6.1 Operations

The system shall operate as a web-based application accessible through modern web browsers.

No client-side installation shall be required.

System maintenance activities (e.g., database backup, server updates) shall be performed during scheduled maintenance windows.

User progress and scores shall be automatically saved after each completed activity.

The system shall log critical errors for monitoring and troubleshooting purposes.

3.6.2 Site Adaptation

The system shall support deployment in cloud-based hosting environments.

Environment-specific configurations (e.g., API keys, database connections) shall be configurable.

The system shall support localization and user interface adaptation for multiple languages.

The architecture shall allow future extensions such as mobile applications or LMS integration without major redesign.

APPENDIX A: Sample Input/Output Formats

This appendix describes representative input and output data formats used by the system to support user interactions. These samples are provided for clarification purposes only and do not represent implementation-level specifications.

A.1 Writing Exercise Submission Input

When a user submits a writing exercise, the system receives the following logical input data:

- User identifier
- Lesson identifier
- Writing exercise type (character, sentence, or paragraph)
- User-written content

The input data is validated to ensure completeness and correctness before evaluation.

A.2 Writing Exercise Evaluation Output

After evaluation, the system returns the following output information:

- Writing score
- Qualitative feedback on writing accuracy
- Submission status (successful or failed)

This output is displayed to the user and stored for progress tracking.

APPENDIX B: API Handshake Pack

This appendix provides a high-level overview of the system's API communication principles. It does not define technical API specifications.

B.1 General API Communication

- The system uses client–server communication to exchange learning data.
- All requests are authenticated to ensure data security.
- Responses follow a consistent structure to simplify client-side handling.

B.2 Leaderboard Data Exchange

- The client requests ranking data from the server.
- The server returns sorted leaderboard information, including user rank and total points.
- Leaderboard data is read-only from the client perspective.

B.3 Online Speaking Room Communication

- The client sends requests to create or join a speaking room.
- The system establishes real-time audio communication between participants.
- Connection status updates are communicated to the client during the session.

APPENDIX C: Error Schema

All API errors returned by the Korean learning web system shall follow a consistent and structured error schema to ensure clear communication between the backend and the client application.

Field	Description
errorCode	A stable, machine-readable error identifier used for error handling logic (e.g., `AUTH_INVALID`, `SPEECH_SERVICE_UNAVAILABLE`).
message	A human-readable error message intended for display in the user interface.
details	An optional list providing additional context or field-level error information.

Common Error Codes

Error Code	Description
AUTH_INVALID	Invalid email or password.
AUTH_UNAUTHORIZED	User is not authenticated. Please log in to continue.
COURSE_NOT_FOUND	The selected course does not exist.

LESSON_NOT_FOUND	The selected lesson does not exist.
LESSON_NOT_COMPLETED	All required lesson sections must be completed before review.
ANSWER_SUBMISSION_FAILED	Failed to submit the user's answer. Please try again.
DATA_RETRIEVAL_ERROR	Unable to retrieve data from the system.
LEADERBOARD_NOT_AVAILABLE	Leaderboard data is not available at the moment.
ROOM_FULL	The selected speaking room has reached its maximum capacity.
ROOM_NOT_FOUND	The selected speaking room does not exist.
CONNECTION_FAILED	Unable to establish a real-time connection. Please retry.
MIC_PERMISSION_DENIED	Microphone access is required for speaking activities.

APPENDIX D: Traceability Matrix

Functional Requirement (FR)	Related Use Case(s)	Related NFR(s)
FR-01 User Authentication	UC-01 Login, UC-02 Register, UC-03 Log out	—
FR-02 Course Browsing & Selection	UC-04 View Courses, UC-05 Select Course	NFR-01, NFR-02
FR-03 Lesson Learning Flow	UC-06 Study Lesson	NFR-01
FR-04 Theory Content Display	UC-07 Learn Theory	NFR-01, NFR-02
FR-05 Reading Practice	UC-08 Practice Reading	NFR-01, NFR-02
FR-06 Listening Practice	UC-09 Practice Listening	NFR-01
FR-07 Speaking Practice	UC-10 Practice Speaking	NFR-01
FR-08 Writing Practice	UC-11 Practice Writing	NFR-01, NFR-02
FR-09 Lesson Review & Scoring	UC-11 Review Lesson	NFR-01
FR-10 Leaderboard & Ranking	UC-12 View Leaderboard	NFR-01, NFR-03
FR-11 Online Speaking Room	UC-13 Participate in Speaking Room	NFR-01, NFR-03

APPENDIX E: Requirements Ownership

This table allocates responsibility for drafting and verifying requirements to ensure coverage by all members. Final allocation can be adjusted during planning.

Member	FRs	NFRs
Bùi Việt Anh	FR-9, FR-10	NFR-01
Nguyễn Nam Anh	FR-01, FR-02, FR-03	
Lã Hoàng Khánh Dương	FR-05, FR-06	
Vũ Hoàng Hà	FR-04, FR-11	NFR-02
Đặng Văn Quân	FR-07, FR-08	NFR-03

APPENDIX F: Course Topic Coverage

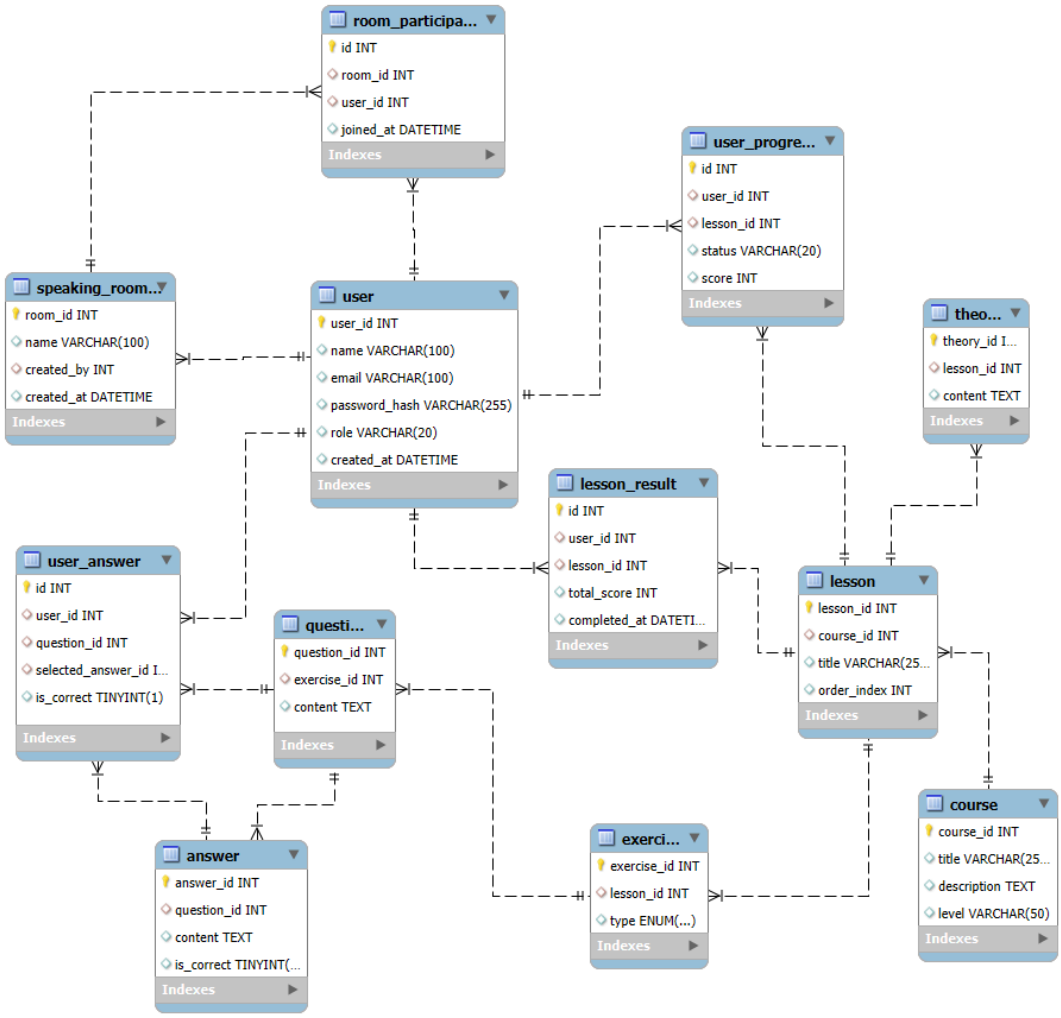
This matrix demonstrates how the project requirements and design artifacts align with the core topics covered in the course. It provides evidence that the project applies theoretical concepts taught in class to a practical software system.

Course Topic	Project Coverage	Evidence / Deliverable
Software Requirements Engineering	The system requirements are clearly identified, analyzed, and documented using a structured SRS format.	Chapter 1 (Introduction), Chapter 2 (General Description), Chapter 3 (Requirements Specification)

Stakeholder & User Analysis	Identification of target users, their characteristics, and usage assumptions.	Section 2.3 User Characteristics
Functional Requirements Specification	Detailed definition of system functions using use-case-based structure.	Section 3.2 Functional Requirements (UC-01 to UC-13)
Non-Functional Requirements	Specification of quality attributes such as performance, usability, and availability.	Section 3.3 Performance Requirements, NFR-01 to NFR-03
Use Case Modeling	Functionalities are decomposed into individual use cases representing user–system interactions.	Section 3.2.1 – 3.2.12 (Use Case Descriptions)
User Interface Design Principles	UI requirements emphasize usability, clarity, and localization support.	Section 3.1.1 User Interfaces, NFR-02
System Performance & Availability	Performance constraints and uptime requirements are defined to ensure system reliability.	Section 3.3 Performance Requirements, NFR-01, NFR-03
Error Handling & Robustness	Standardized API error schema ensures consistent error reporting and system robustness.	APPENDIX C: Error Schema
Requirements Traceability	Mapping between functional requirements, use cases, and non-functional requirements.	APPENDIX D: Traceability Matrix

Team Collaboration & Responsibility Allocation	Clear ownership of functional and non-functional requirements among team members.	APPENDIX E: Requirements Ownership
Web-based Application Architecture	The system is designed as a web-based learning platform with defined interfaces and dependencies.	Sections 2.1 Product Perspective, 3.1 External Interface Requirements

APPENDIX G: Database Design



Entity Relationship Diagram (ERD)

Entity	Attribute	Input Type	Key
User	user_id	INT (AUTO_INCREMENT)	PK
	name	VARCHAR(100)	
	email	VARCHAR(100)	
	password_hash	VARCHAR(255)	
	role	VARCHAR(20)	
	created_at	DATETIME	
Course	course_id	INT (AUTO_INCREMENT)	PK
	title	VARCHAR(255)	
	description	TEXT	
	level	VARCHAR(50)	
Lesson	lesson_id	INT (AUTO_INCREMENT)	PK
	course_id	INT	FK
	title	VARCHAR(255)	
	order_index	INT	
Theory	theory_id	INT (AUTO_INCREMENT)	PK
	lesson_id	INT	FK
	content	TEXT	
Exercise	exercise_id	INT (AUTO_INCREMENT)	PK
	lesson_id	INT	FK
	type	ENUM('listening','speaking','reading','writing')	
Question	question_id	INT (AUTO_INCREMENT)	PK
	exercise_id	INT	FK
	content	TEXT	

Answer	answer_id	INT (AUTO_INCREMENT)	PK
	question_id	INT	FK
	content	TEXT	
	is_correct	BOOLEAN	
User_Answer	id	INT (AUTO_INCREMENT)	PK
	user_id	INT	FK
	question_id	INT	FK
	selected_answer_id	INT	FK
	is_correct	BOOLEAN	
User_Progress	id	INT (AUTO_INCREMENT)	PK
	user_id	INT	FK
	lesson_id	INT	FK
	status	VARCHAR(20)	
	score	INT	
Lesson_Result	id	INT (AUTO_INCREMENT)	PK
	user_id	INT	FK
	lesson_id	INT	FK
	total_score	INT	
	completed_at	DATETIME	
Speaking_Room	room_id	INT (AUTO_INCREMENT)	PK
	name	VARCHAR(100)	
	created_by	INT	FK
	created_at	DATETIME	
	id	INT (AUTO_INCREMENT)	PK
Room_Participant	room_id	INT	FK

	user_id	INT	FK
	joined_at	DATETIME	

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